RoboPIN

Grounded Embodied Reasoning via Pinned Chain-of-Thought

Yaoting Huang* · Yifu Yuan*† · Linqi Han · Chengwen Li · Shuoheng Zhang
Xianze Yao · Hongyao Tang · Yan Zheng · Jianye Hao†

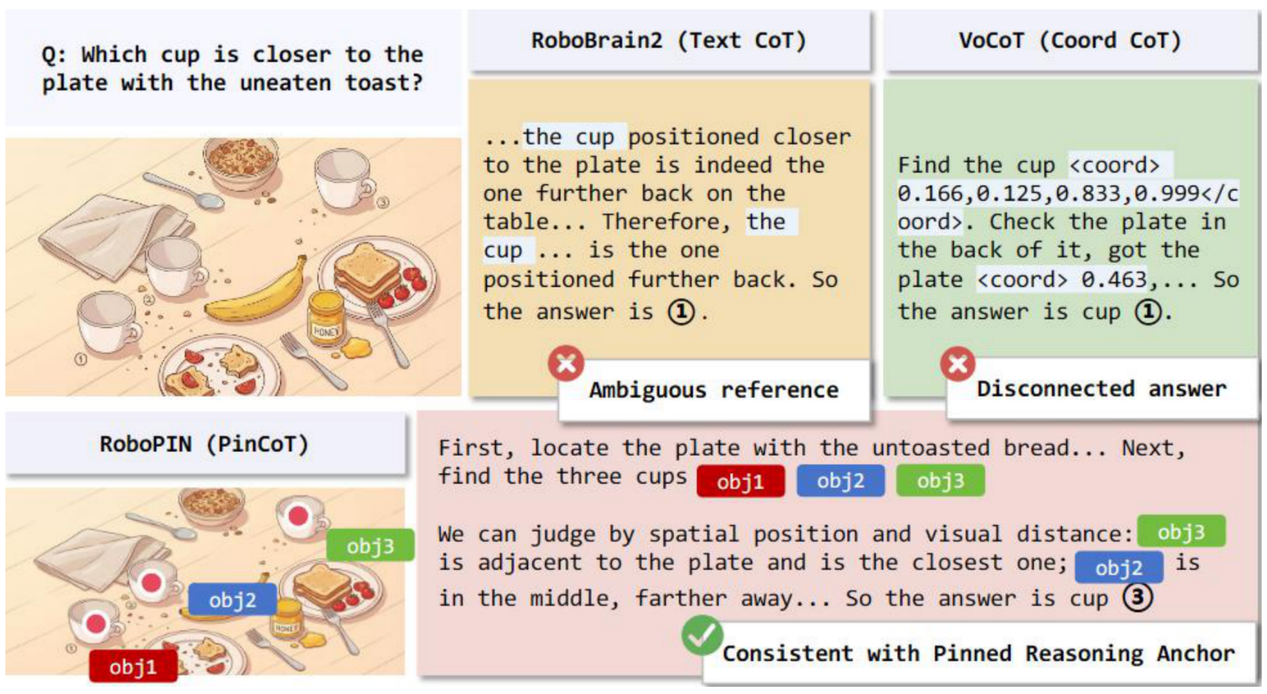
Tianjin University · ACM Multimedia 2026 · Rio de Janeiro, Brazil

quark404@tju.edu.cn · yuanyf@tju.edu.cn | * Equal contribution † Corresponding authors

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Tianjin University

Motivation



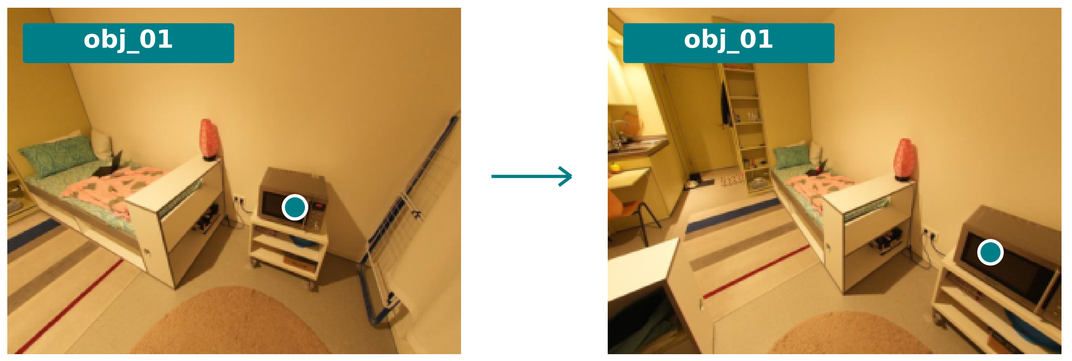
1 Reference drift

Textual references can switch between similar objects during multi-step reasoning.

2 Reasoning–answer misalignment

A correct intermediate analysis can still end on the wrong object, especially across views.

Key Insight: Persistent Identity



Same entity. Same ID. Different views.

Reasoning anchor = name + ID + view + point.

Introduce the anchor; reuse its identity throughout the reasoning chain.

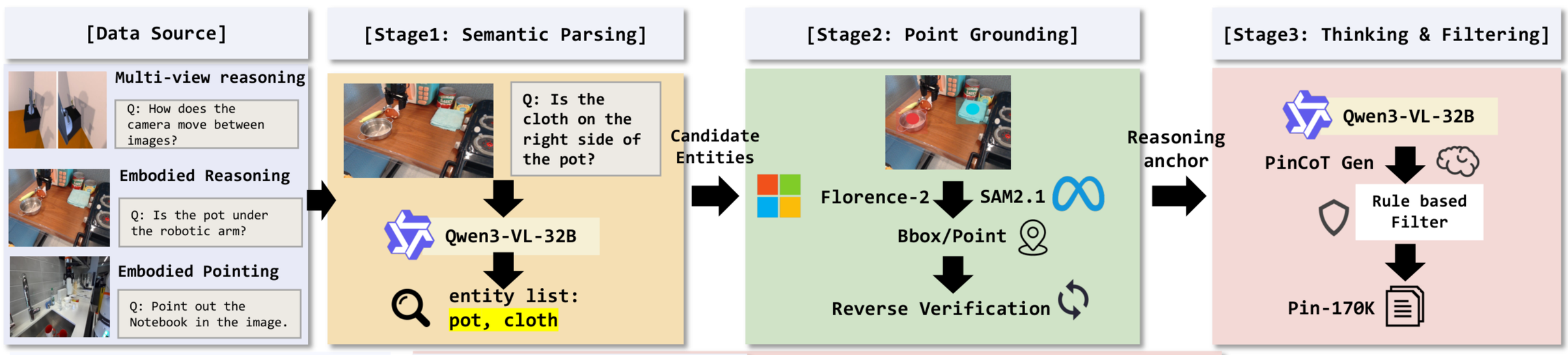
Our insight

Persistent IDs connect visual grounding, intermediate steps and the final answer.

QR PLACEHOLDER

PAPER / CODE

Our Method: RoboPIN / PinCoT

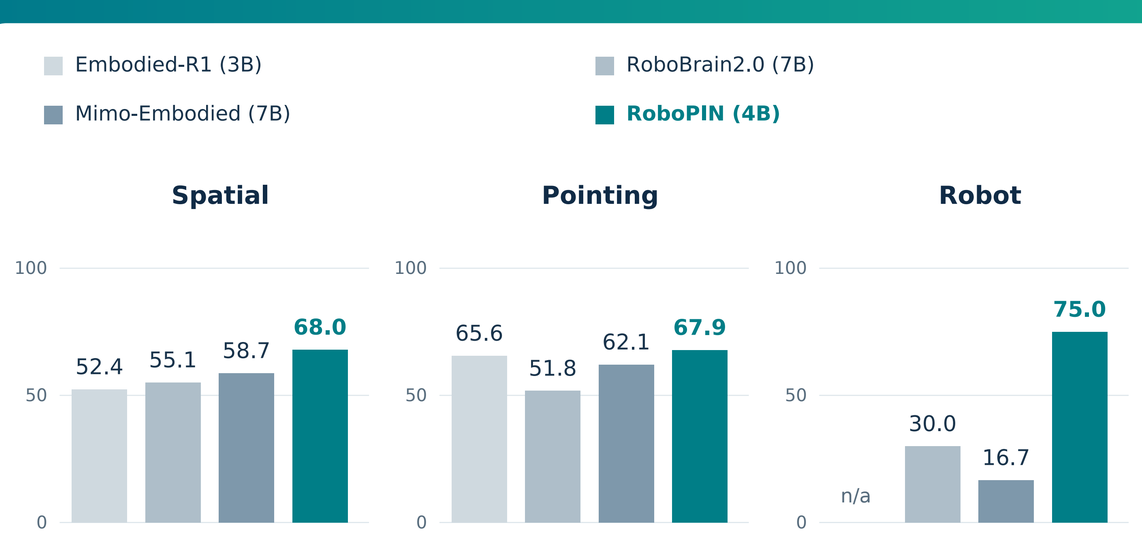


Three-stage post-training

SFT — Embodied knowledge → **CoT-SFT** — Structured reasoning → **RFT / GRPO** — Process alignment

PinCoT binds visual evidence to persistent identities. PIN-170K supplies structured traces; process rewards supervise grounding and identity consistency.

Experimental Results



Scores (%) · Spatial: 13-column mean; pointing: 5-benchmark mean; robot: success over 60 trials.

18.8% → 99.9% cross-step ID coverage

Without RFT → RoboPIN · EmbSpatial

Qualitative Results

Zero-shot xArm

Spatial
20 / 20

Multi-view
14 / 20

Long-horizon
11 / 20

Successes per task

Persistent visual anchors make embodied reasoning traceable.